

Artifact Evaluation Report: VLASelect

1. Environment and Execution setup

1.1 Platform Specifications

Table 1.1: Hardware and Software Configuration Comparison.

Subsystem	Platform for Paper (Full Run)	Platform for MWE
Operating System	Ubuntu 20.04/22.04 LTS (Kernel 5.4/5.15)	Ubuntu 22.04.4 LTS (Kernel 6.8)
CPU Architecture	2 × Intel Xeon Gold 6430 (64C)	Intel Xeon E5-2698 v4 (16C)
System Memory	128 GB DDR5	32 GB DDR4
GPU & VRAM	NVIDIA A100 80GB (+ Edge AGX)	NVIDIA Tesla V100 (32 GB)
CUDA Toolchain	Driver 550.144.03, CUDA 12.4	Driver 550.127.05, CUDA 12.4

1.2 Execution Workflow Integrity

Minimal Working Example (MWE=1): The test script is verified and executed, key steps include initialization, forward propagation, neuron importance scoring, and backward updates.

Full-Scale Example: Executed reinforcement learning iterations in standardized ManiSkill environments to assess dynamic policy adaptation.

1. Evaluation Reproduction

To evaluate the extended deployment capabilities of VLASelect, we performed assessments across sim-to-real transfer, ICL comparison, and multi-agent scenarios.

2.1 (Figure 7) Accuracy Under Tasks/Environment Changes

To evaluate the model's accuracy under varying task and environment conditions, we tested VLASelect across four distinct workloads; both full and minimal runs (as shown in Figure 2.1(a) and Figure 2.1(b)) **achieved the best overall accuracy**. The minimal test evaluated a subset of models with shorter runtime, while the full test required longer execution, and both confirmed our method's superior performance.

Key observation: VLASelect consistently performs **Highest average accuracy** under tasks and environment changes.

For experiments in this paper, the runtime is ~ 140 hours, peak VRAM footprint is 60 GB.

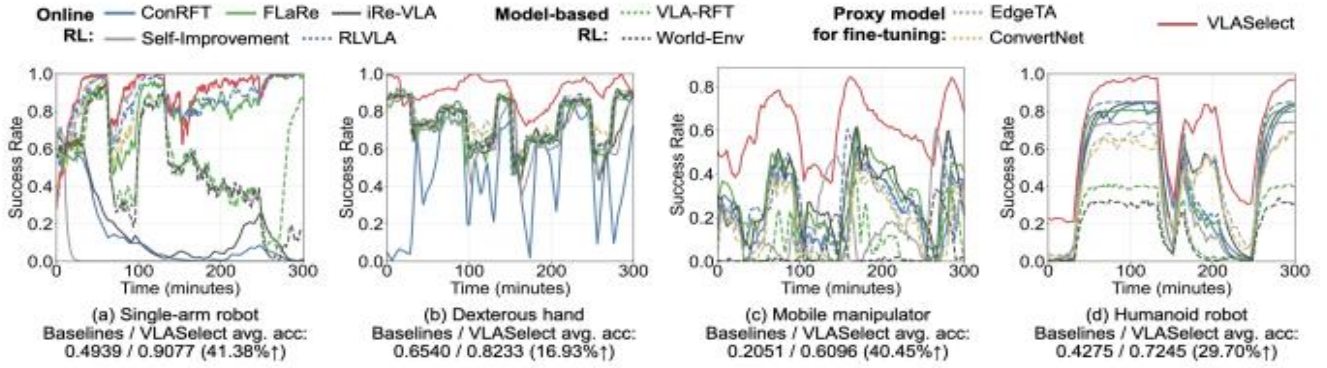


Figure 2.1(a): Full-scale Example for Accuracy Under Task and Environment Changes

For minimal working example, the runtime is ~ 1 hour, peak VRAM footprint is 20 GB.

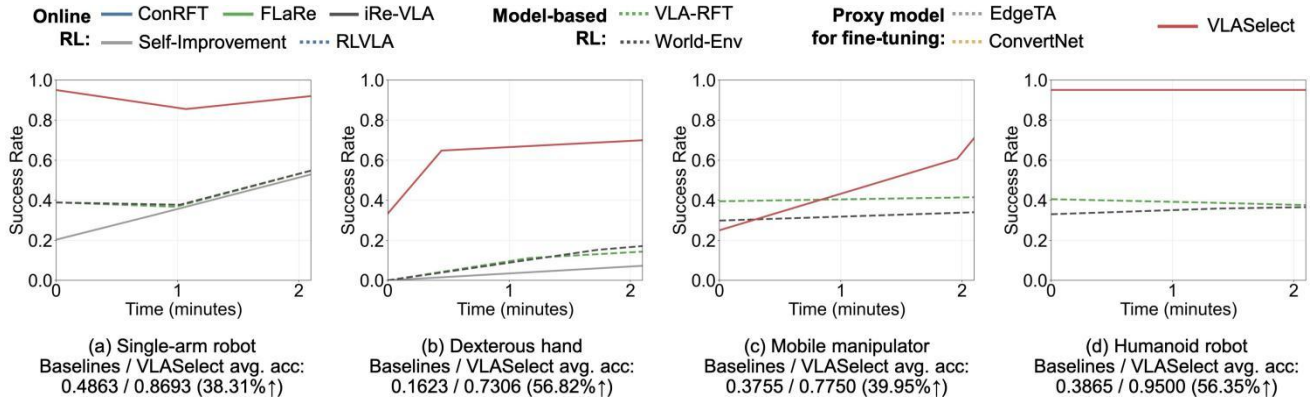


Figure 2.1(b): Minimal Working Example for Accuracy Under Task and Environment Changes

2.2 (Figure 8) Accuracy Under Available Resource Changes

To evaluate VLASelect's performance under fluctuating resource availability, this section employs the same contrastive configuration as described above; results illustrate that **VLASelect consistently performs best under these conditions**. The full run (Figure 2.2a) and the minimized run (Figure 2.2b) both confirm the model's superior overall accuracy.

Key observation: VLASelect consistently achieves the **highest overall accuracy** under fluctuating resource availability.

For experiments in this paper, the runtime is ~ 140 hours, peak VRAM footprint is 60 GB.

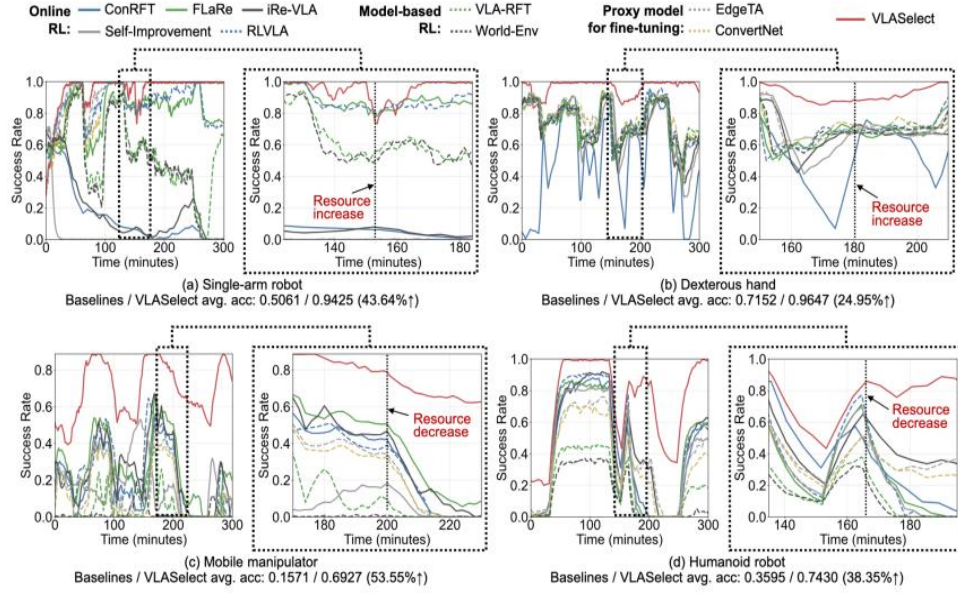


Figure 2.2(a): Full-scale Example for Accuracy Under Available Resource Changes

For minimal working example, the runtime is ~ 1 hour, peak VRAM footprint is 20 GB.

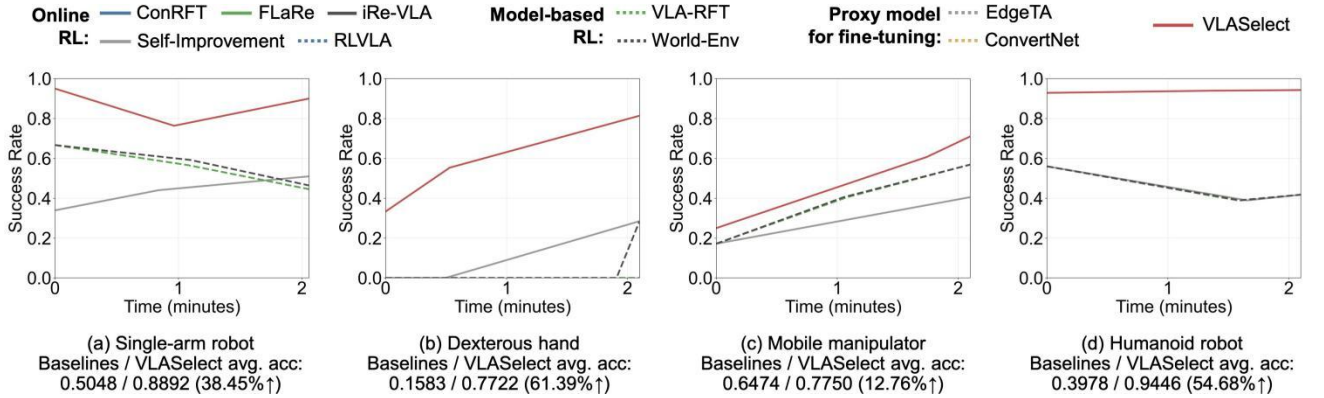


Figure 2.2(b): Minimal Working Example for Accuracy Under Available Resource Changes

2.3 (Figure 9 and Tables 2/3) Overheads Under The Same Accuracy

Across both full scale (Figure 2.3(a)) and minimized (Figure 2.3(b)) experiment configurations, VLASelect consistently **maintains a lower peak VRAM footprint** than all baselines during online RL. Crucially, **it attains comparable task accuracy in the shortest Reinforcement learning time**, enabling significantly earlier resource release.

Key observation: VLASelect achieves comparable task accuracy with the **lowest peak VRAM footprint and the shortest reinforcement learning time** during online RL.

For experiments in this paper, the runtime is ~ 140 hours, peak VRAM footprint is 60 GB.

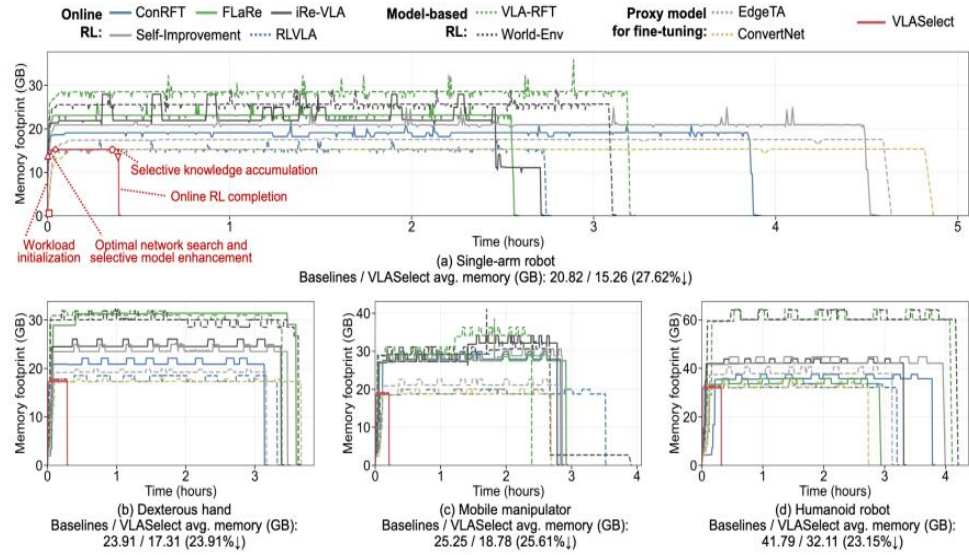


Figure 2.3(a): Full-scale Example for Memory Footprint Comparison in a new task

For minimal working example, the runtime is ~ 1 hour, peak VRAM footprint is 20 GB.

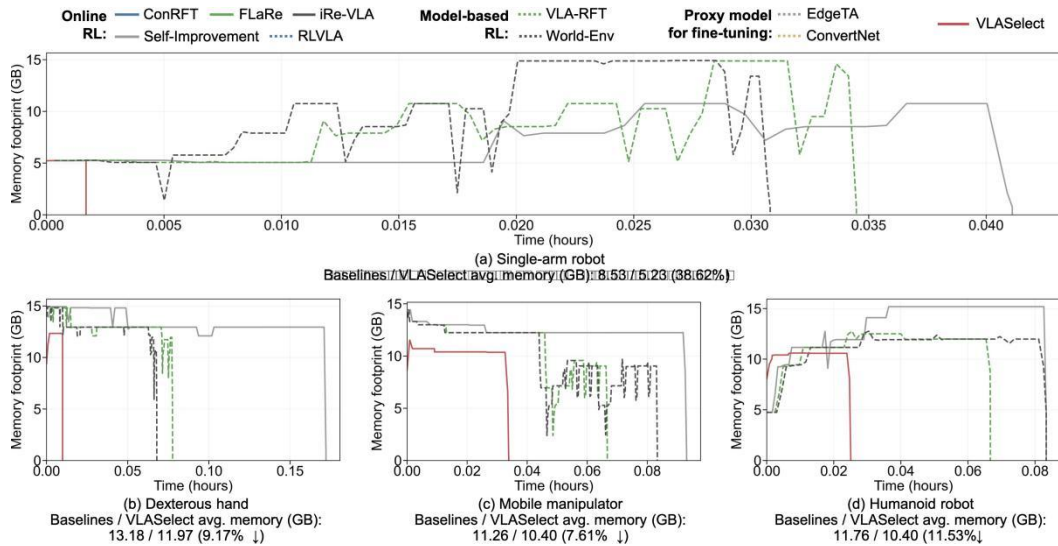


Figure 2.3(b): Minimal Working Example for Memory footprint

Energy Consumption Analysis

Tables 2.3(a) and 2.3(b) compare the average energy consumption (kJ) across baseline methods under the full-scale and minimized experimental settings. Across all robotic workloads in both configurations, **VLASelect**

consistently incurs the lowest energy consumption, demonstrating superior operational efficiency during online adaptation.

Key observation: VLASelect achieves the highest operational efficiency by maintaining the lowest energy consumption across diverse workloads and experimental settings.

For experiments in this paper, this table generated concurrently with the previous step.

Table 2.3(a): Full-scale Example for Average Energy Consumption (kJ) in each new task

Method	Single-arm robot	Dexterous hand	Mobile manipulator	Humanoid robot
ConRFT	342.37	346.31	499.84	787.19
	403.51	274.05	291.68	570.22
FlaRe	318.06	477.07	461.29	671.17
	254.18	317.51	379.37	452.06
iRe-VLA	383.33	530.53	505.4	661.88
	213.81	204.04	273.55	564.21
Self-Improvement	551.76	416.37	497.49	801.52
	473.14	293.83	297.81	673.49
RLVLA	326.26	360.71	573.44	622.44
	206.67	314.32	466.58	458.46
VLA-RFT	323.69	433.42	374.09	795.6
	277.8	280.17	257.39	632.21
World-Env	329.26	339.87	666.9	775.05
	265.45	451.14	480.57	639.55
EdgeTA	349.4	383.21	464.28	629.97
	549.99	282.31	355.5	444.5
ConvertNet	608.37	456.51	451.61	536.37
	454.23	327.35	378.12	451.6
VLASelect	46.06	35.41	39.46	64.71
	44.24	25.63	25.57	45.18

For minimal working example, this table generated concurrently with the previous step.

Table 2.3(b): Minimal Working Example for Average Energy Consumption (kJ) in each new task

Method	Single-arm robot	Dexterous hand	Mobile manipulator	Humanoid robot
Self-Improvement	15.82	62.99	37.17	26.22
VLA-RFT	13.83	27.52	26.83	24.62
World-Env	12.98	23.96	33.92	33.03
VLASelect	0.68	11.37	12.28	2.58

Overhead comparison under the same learning accuracy

To evaluate system overhead under same accuracy conditions, we measure the computational cost required to reach target learning performance. Across both the full-scale (Table 2.3(c)) and minimized (Table 2.3(d))

experimental settings, **VLASelect consistently achieves the target accuracy with shortest training time and reduced resource consumption** compared to all baseline methods.

Key observation: Under identical target accuracy, VLASelect minimizes computational overhead by **achieving the shortest learning time and lowest resource consumption**.

For experiments in this paper, this table generated concurrently with the previous step.

Table 2.3(c): Full-scale Example for Overhead comparison under the same learning accuracy

	Time (h)				Memory footprint (GB)				Energy (kJ)			
Method	Single-arm robot	Dexterous hand	Mobile manipulator	Humanoid robot	Single-arm robot	Dexterous hand	Mobile manipulator	Humanoid robot	Single-arm robot	Dexterous hand	Mobile manipulator	Humanoid robot
ConRFT	3.88	3.14	2.83	3.78	19.12	20.88	27.92	35.71	360.39	294.89	334.87	600.23
FlaRe	2.56	3.6	2.92	2.91	23.14	31.39	27.49	33.7	295.13	384.55	384.75	492.3
iRe-VLA	2.71	3.38	2.78	3.31	21.93	24.55	27.3	42.19	253.73	374	382.41	630.36
Self-Improvement	4.52	3.48	2.86	3.95	20.93	23.52	27.53	42.11	547.71	309.29	337.9	763.35
RLVLA	2.74	3.32	3.52	3.2	15.26	17.31	18.78	32.11	275.99	353.41	546.13	608.56
VLA-RFT	3.2	3.39	2.39	4.07	28.54	30.99	29.04	60.02	319.56	412.78	341.89	704.55
World-Env	3.1	3.63	3.87	4.16	25.66	30.05	29.27	60.23	279.42	429.66	505.86	738.14
EdgeTA	4.59	3.17	2.84	3.12	17.54	19.23	21.12	37.87	523.8	382.06	440.83	467.9
Convert Net	4.82	3.67	2.69	2.73	15.26	17.31	18.78	32.11	478.14	434.77	430.1	510.83
VLASelect	0.39	0.29	0.21	0.32	15.26	17.31	18.78	32.11	45.53	34.53	28.76	47.56

For minimal working example, this table generated concurrently with the previous step.

Table 2.3(d): Minimal Working Example for Overhead comparison under the same learning accuracy

	Time (h)				Memory footprint (GB)				Energy (kJ)			
Method	Single-arm robot	Dexterous hand	Mobile manipulator	Humanoid robot	Single-arm robot	Dexterous hand	Mobile manipulator	Humanoid robot	Single-arm robot	Dexterous hand	Mobile manipulator	Humanoid robot
Self-Improvement	0.04	0.17	0.09	0.08	7.32	13.41	12.47	12.94	15.82	62.99	37.17	25.65
VLA-RFT	0.03	0.08	0.07	0.07	8.48	13.09	11.07	11.21	13.83	27.52	26.83	25.24
World-Env	0.03	0.07	0.08	0.08	9.77	13.03	10.25	11.12	12.98	23.96	33.92	33.02
VLASelect	0.01	0.01	0.03	0.03	5.23	11.97	10.4	10.4	0.58	2.96	12.28	9.91

2.4 (Figure 10) Time Breakdown of VLASelect's Modules

To evaluate the computational efficiency of the proposed framework, we analyze the execution time of VLASelect core modules; Figure 2.4 demonstrates that **execution time of VLASelect is obviously less than training iteration time**.

Key observation: The execution time across all VLASelect modules is significantly **shorter than the training iteration time**

For experiments in this paper, the runtime is ~20 minuets, peak VRAM footprint is 60 GB.

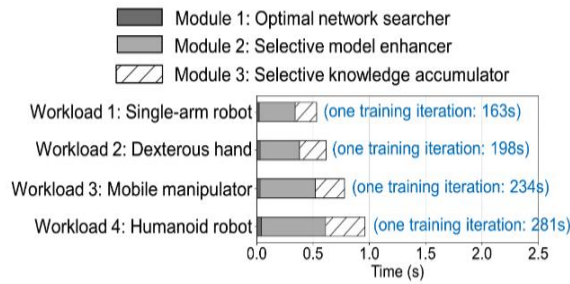


Figure 2.4(a): Full-scale Example of Time Breakdown Experiment

For minimal working example, the runtime is ~20 minuets, peak VRAM footprint is 20 GB.

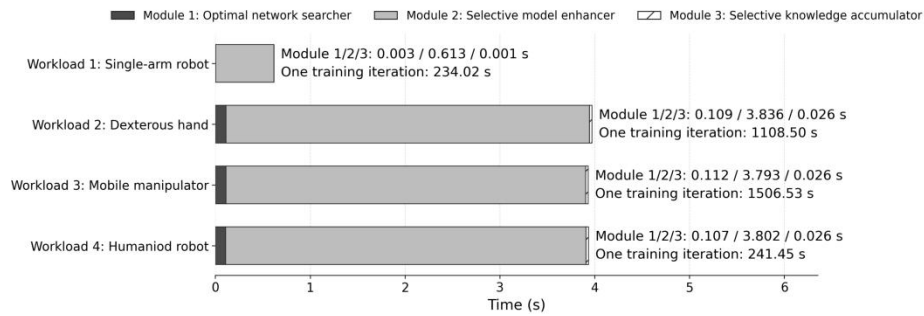


Figure 2.4(b): Minimal Working Example of Time Breakdown Experiment

2.5 (Figure 11) Training Time Breakdown in Each Workload

To evaluate and compare the efficiency of sampling and training across models, we analyzed the time breakdown for full-scale and minimized experiments (Figure 2.5(b)), showing that under identical workloads, **VLASelect achieves shorter runtime** while maintaining high accuracy compared with other baselines.

Key observation: Under identical workloads, VLASelect consistently **achieves shortest runtime** across sampling and training while maintaining high accuracy.

For experiments in this paper, the runtime is ~140 hours, peak VRAM footprint is 60 GB.

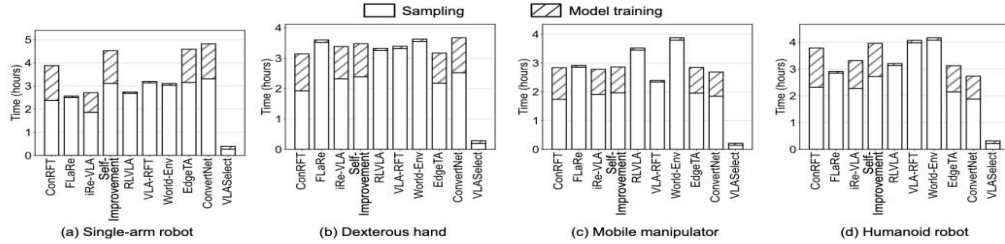


Figure 2.5(a): Full-scale Example for Time Breakdown of all models

For minimal working example, the runtime is ~1 hour, peak VRAM footprint is 20 GB.

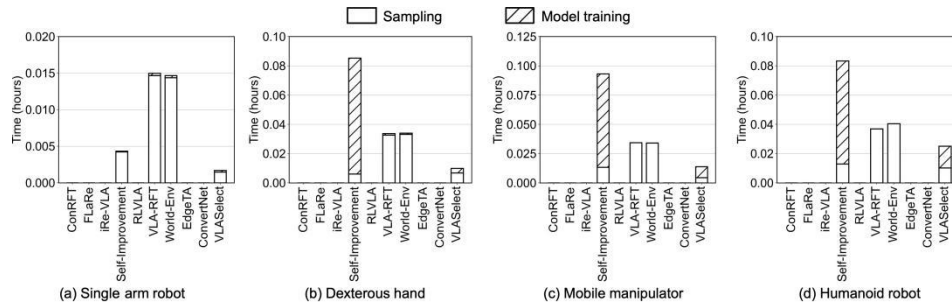


Figure 2.5(a): Minimal Working Example of Time Breakdown for all models

2.6 (Figure 12) Design Choice Validation by Ablation

To evaluate the individual contribution of each module to overall performance, we conducted an ablation study; under both configurations (Figure 2.6(a) and Figure 2.6(b)), **all modules consistently contribute to the final accuracy**.

Key observation: All individual modules in VLASelect consistently contribute to the overall task accuracy across both configurations.

For experiments in this paper, the runtime is ~40 hours, peak VRAM footprint is 60 GB.

For minimal working example, the runtime is ~1 hour, peak VRAM footprint is 20 GB.

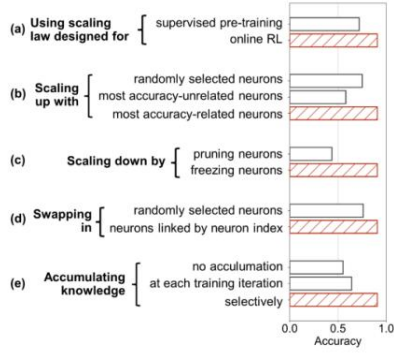


Figure 2.6(a): Full-scale Example for Ablation Experiment

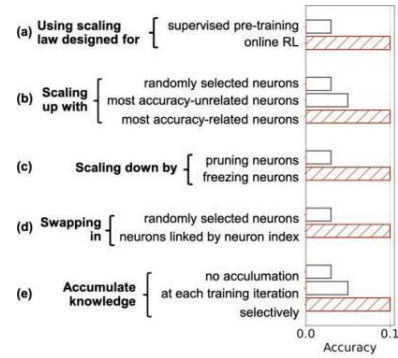


Figure 2.6(b): Minimal Working Example for Ablation Verification

2.7 Discussion 1: Sim-to-Real Transfer

A supplementary video will be provided to compare simulation and real practice, further verifying the method's consistency and generalization across virtual and real environments.

2.8 Discussion 2: ICL (In-Context Learning)

In the full-scale evaluation, **VLASelect achieves a higher final accuracy than RICL (49.74% vs. 38.13%)**. In the minimal working example (~1.6 minutes), **RICL fails entirely** because the brief runtime is insufficient for its policy to update, whereas **VLASelect rapidly adapts and maintains an average accuracy of 37.2%**.

Key observation: VLASelect outperforms RICL in final accuracy and maintains rapid adaptation.

2.9 Discussion 3: Maximum Supported Model Size

This part shows the maximum supported model size under different configurations. As listed in Table 2.5.4, VLA Select supports up to **11.3 GB on Xavier and 24.0 GB on Orin** in the full-scale test, while the minimal working experiment ranges **from 2.5 GB to 24 GB on V100 (32 GB)**.

Key observation: **VLASelect demonstrates broad hardware adaptability**, supporting maximum model sizes of up to 11.3 GB on Xavier and 24.0 GB on Orin.

For both experiments in this paper and minimal working example, the runtime is ~1 hour, peak VRAM footprint is 32 GB.

Table 2.9 Example of Maximum supported model size

Test platform	Maximum model size
Full run	11.3 GB (Xavier), 24.0 GB (Orin)
MWE run	from 2.5 to 24 GB (V100 32G)

2.10 Discussion 4: Applicability to multi-agent scenarios

In the full-scale experiment, **VLASelect improves average accuracy by 17.3% over MAPPO**. Under the minimized setting (~0.5 minutes), MAPPO completely fails due to the short runtime being insufficient for multi-agent policy updates, whereas **VLASelect reaches up to 60.0% accuracy**.

Key observation: VLASelect surpasses MAPPO by **17.3%** in full-scale evaluation and maintains up to **60.0% accuracy** under minimal working experiment.

For experiments in this paper, the runtime is ~7 hours, peak VRAM footprint is 60 GB.

For minimal working example, the runtime is ~20 minutes, peak VRAM footprint is 20 GB.

2. Supporting Various VLA Models, Scaling Strategies, and Knowledge Exchange Granularities

To evaluate VLASelect's generalization across VLA-Adapter backbones, we performed assessments on continual learning trajectories, scaling strategies, and knowledge exchange angularities.

3.1 Example 1: VLA-Adapter

3.1.1 Supporting the VLA-Adapter

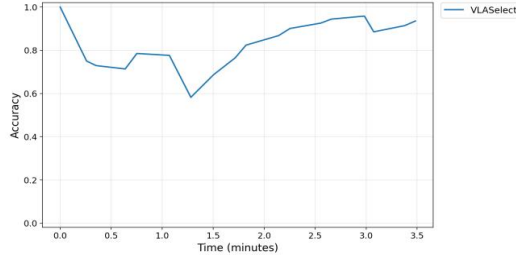


Figure 3.1.1(a): Full scale evaluation

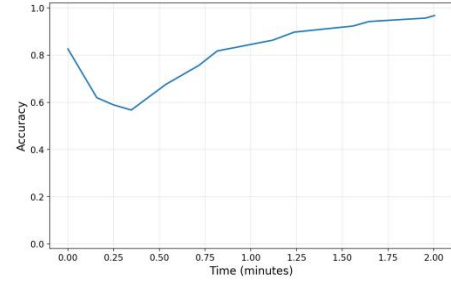


Figure 3.1.1(b): Minimal Working Example

Figure 3.1.1: Online RL Continual Learning Trajectory for VLA-Adapter.

For experiments in this paper, the runtime is ~3 hours, peak VRAM footprint is 60 GB.

For minimal working example, the runtime is ~3 minutes, peak VRAM footprint is 20 GB.

Key observation: VLASelect adapts models through a unified interface via VLA-Adapter and runs training successfully, with consistent result trends across both platforms.

3.1.2 Supporting different scaling strategies

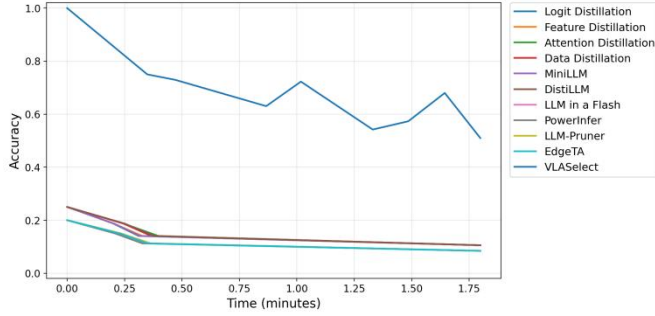


Figure 3.1.2(a): Full scale evaluation

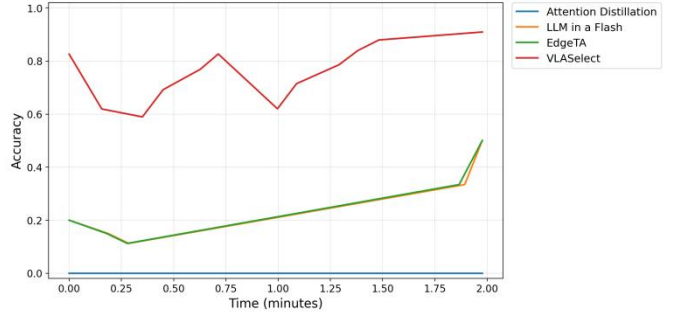


Figure 3.1.2(b): Minimal Working Example

Figure 3.1.2: VLASelect vs. Competing strategies on VLA-Adapter.

For experiments in this paper, the runtime is ~40 hours, peak VRAM footprint is 60 GB.

For minimal working example, the runtime is ~20 minutes, peak VRAM footprint is 20 GB.

Key observation: VLASelect integrates VLA-Adapter and outperforms knowledge distillation and dynamic pruning in selective model scaling.

3.1.3 Supporting different knowledge exchange granularities

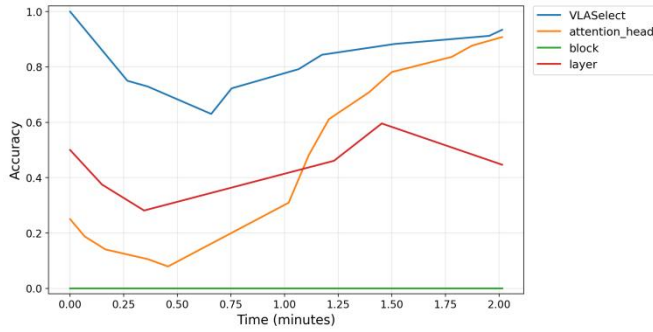


Figure 3.1.3(a): Full scale evaluation

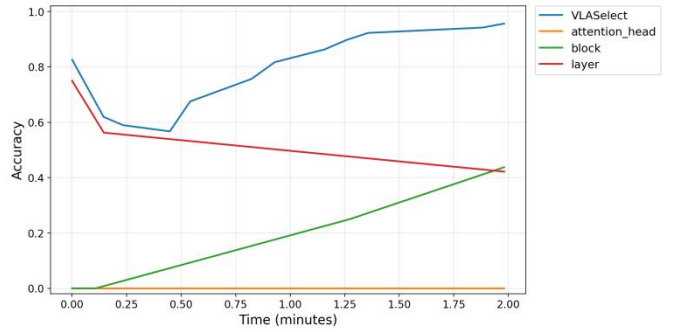


Figure 3.1.3(b): Minimal Working Example

Figure 3.1.3: accuracy for different knowledge exchange granularities on VLA-Adapter

For experiments in this paper, the runtime is ~15 hours, peak VRAM footprint is 60 GB.

For minimal working example, the runtime is ~30 minutes, peak VRAM footprint is 20 GB.

Key observation: VLASelect with VLA-Adapter supports multiple knowledge exchange granularities, with channel/neuron outperforming coarser block, layer, and attention head levels.

3.2 Example 2: TinyVLA

To evaluate the model's performance on the TinyVLA backbone, we conducted experiments on continual learning trajectories and scaling strategies.

3.2.1 Supporting the TinyVLA

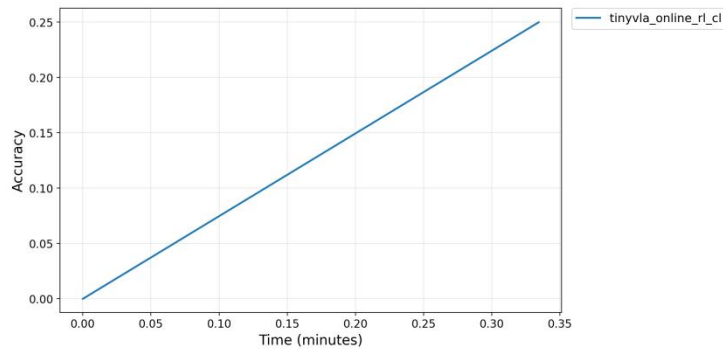


Figure 3.2.1: Minimal Working Example of Online RL Continual Learning Trajectory for TinyVLA

For experiments in this paper, the runtime is ~ 3 hours, peak VRAM footprint is 60 GB.

For minimal working example, the runtime is ~ 3 minutes, peak VRAM footprint is 20 GB.

Key observation: VLASelect supports TinyVLA through a unified adapter interface and trains successfully (only evaluated on constrained testbed).

3.2.2 Supporting different scaling strategies

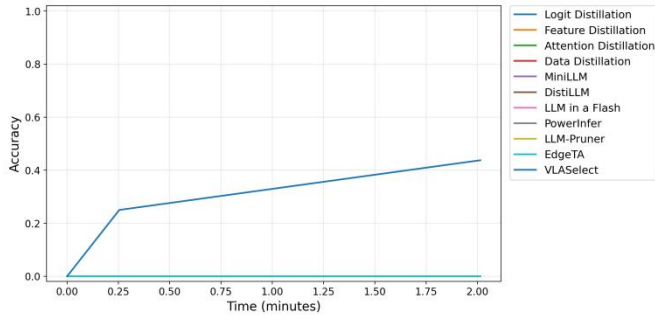


Figure 3.2.2(a): Full scale evaluation

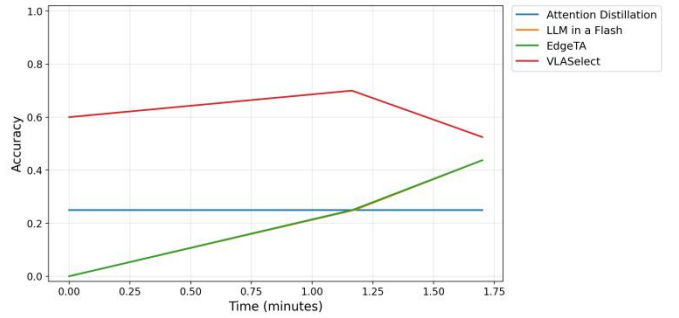


Figure 3.2.2(b): Minimal Working Example

Figure 3.2.2: VLASelect vs. Competing strategies on TinyVLA.

For experiments in this paper, the runtime is ~ 40 hours, peak VRAM footprint is 60 GB.

For minimal working example, the runtime is ~ 20 minutes, peak VRAM footprint is 20 GB.

Key observation: After integrating TinyVLA, VLASelect also combines with knowledge distillation and dynamic pruning, and its selective model scaling still achieves the best results.

3.2.3 Supporting different knowledge exchange granularities

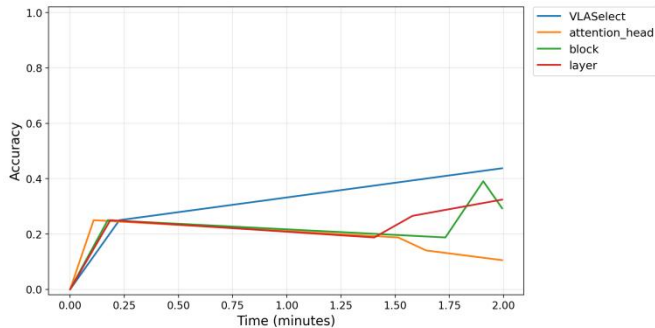


Figure 3.2.3(a): Full scale evaluation

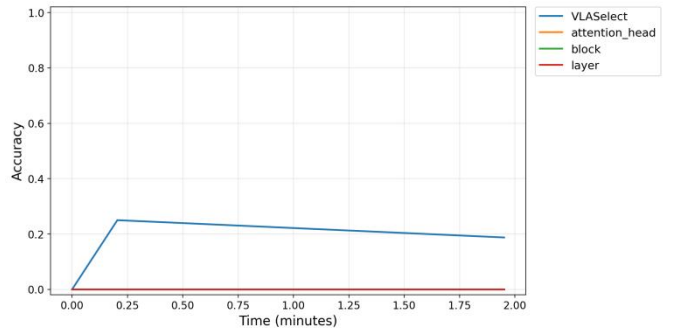


Figure 3.2.3(b): Minimal Working Example

Figure 3.2.3: accuracy for different knowledge exchange granularities on TinyVLA

For experiments in this paper, the runtime is ~15 hours, peak VRAM footprint is 60 GB.

For minimal working example, the runtime is ~30 minutes, peak VRAM footprint is 20 GB.

Key observation: For TinyVLA, VLASelect again performs best with channel/neuron-level knowledge exchange compared to coarser block, layer, and attention head granularities.

3.3 Example 3: EdgeVLA

To evaluate VLASelect's adaptability to the EdgeVLA architecture, we performed benchmarks on online RL trajectories and baseline comparisons.

3.3.1 Supporting the EdgeVLA

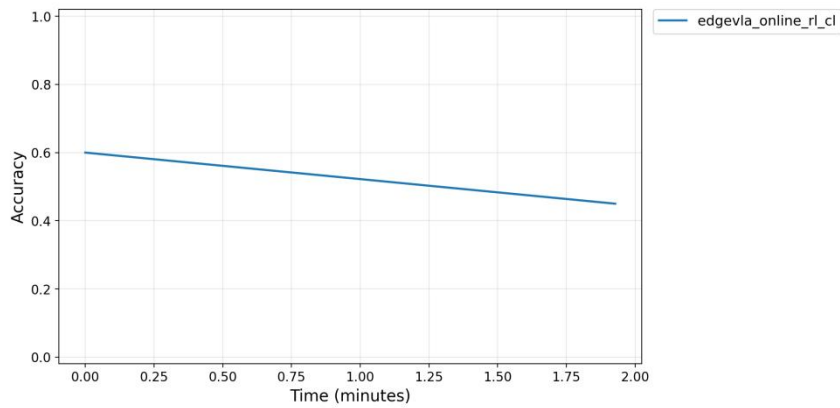


Figure 3.3.1: Minimal Working Example of Online RL Continual Learning Trajectory for EdgeVLA

For experiments in this paper, the runtime is ~3 hours, peak VRAM footprint is 60 GB.

For minimal working example, the runtime is ~3 minutes, peak VRAM footprint is 20 GB.

Key observation: VLASelect adapts EdgeVLA via the same unified interface and trains successfully, with consistent result trends across both platforms (only evaluated on constrained testbed).

3.3.2 Baseline Comparison on EdgeVLA

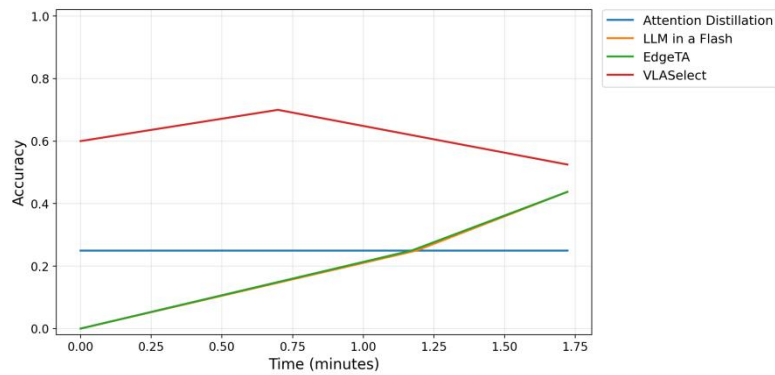


Figure 3.3.2: Minimal Working Example of VLASelect and Baseline Competing strategies on on EdgeVLA

For experiments in this paper, the runtime is ~40 hours, peak VRAM footprint is 60 GB.

For minimal working example, the runtime is ~20 minutes, peak VRAM footprint is 20 GB.

Key observation: Similarly on EdgeVLA, VLASelect outperforms knowledge distillation and dynamic pruning with its selective model scaling approach (only evaluated on constrained testbed).

3.3.3 Swapping Granularity Ablation on EdgeVLA

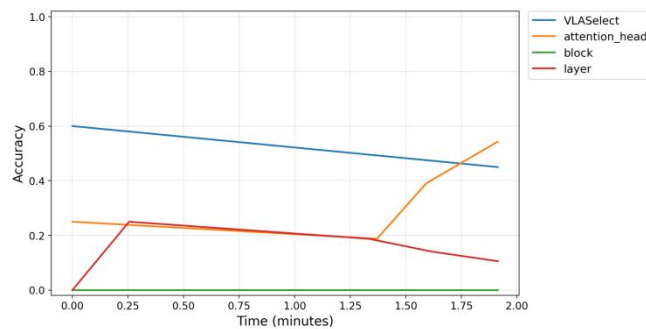


Figure 3.3.3: Minimal Working Example of accuracy for different knowledge exchange granularities on EdgeVLA

For experiments in this paper, the runtime is ~15 hours, peak VRAM footprint is 60 GB.

For minimal working example, the runtime is ~30 minutes, peak VRAM footprint is 20 GB.

Key observation: On EdgeVLA, channel/neuron-level knowledge exchange again yields the best results, surpassing coarser block, layer, and attention head granularities (only evaluated on constrained testbed).